BASIC EPIDEMIOLOGY

2nd Edition

COMPREHENSIVE SUMMARY

Original Authors: R. Bonita, R. Beaglehole, T. Kjellstrom

World Health Organization, 2006

This document provides a comprehensive 25-page summary of the 226-page WHO publication "Basic Epidemiology" (2nd Edition). It preserves all key concepts, definitions, methods, formulas, and essential examples from the original text across all 11 chapters.

Summary prepared: June 2026

Introduction

Epidemiology is the study of the distribution and determinants of health-related states or events in specified populations, and the application of this study to the prevention and control of health problems. This textbook provides an introduction to the basic principles and methods of epidemiology for professionals in health and environment fields.

The purpose of this book is to: explain principles of disease causation with emphasis on modifiable environmental factors; encourage application of epidemiology to disease prevention and health promotion; prepare health professionals for addressing population health; and encourage good clinical practice through clinical epidemiology concepts.

Learning objectives include: understanding the nature and uses of epidemiology; the epidemiological approach to defining and measuring health-related states; strengths and limitations of study designs; the approach to causation; contributions to prevention, health policy, and clinical practice; and evaluating effectiveness and efficiency of health care.

Chapter 1: What is Epidemiology?

Key Messages

- Epidemiology is a fundamental science of public health.
- Epidemiology has made major contributions to improving population health.
- Epidemiology is essential to identifying and mapping emerging diseases.
- There is often a frustrating delay between acquiring epidemiological evidence and applying it to health policy.

Historical Context

Epidemiology originates from Hippocrates' observation over 2000 years ago that environmental factors influence disease occurrence. John Snow's work in London (1848-54) linking cholera to contaminated water supply is a landmark example. He showed that cholera death rates were 5.0 per 1000 in the district supplied by the Southwark company vs. 0.9 per 1000 supplied by the Lambeth company. Snow's work led to public health improvements decades before the causative organism was identified.

Richard Doll and Andrew Hill established the association between smoking and lung cancer through long-term cohort studies beginning in the 1950s. Male doctors born 1900-1930 who smoked died on average 10 years younger than lifelong non-smokers.

Definition, Scope, and Uses

Definition (Last): "The study of the distribution and determinants of health-related states or events in specified populations, and the application of this study to the prevention and control of health problems."

The word derives from Greek: epi (upon) + demos (people) + logos (study). Epidemiology encompasses: Study (surveillance, observation, hypothesis testing, experiments); Distribution (time, persons, places); Determinants (biological, chemical, physical, social, cultural, economic, genetic, behavioural factors); Health-related states (diseases, causes of death, behaviours, positive health states); and Application to prevention and control.

Uses of Epidemiology

- Causation of disease: Most diseases result from interaction between genetic and environmental factors.
- Natural history of disease: Course and outcome in individuals and groups.
- Health status of populations: Describing disease burden for health planning.
- Evaluating interventions: Determining effectiveness and efficiency of health services.

Achievements

Key achievements include: Smallpox eradication (declared 8 May 1980); identification of methyl mercury poisoning in Minamata, Japan; understanding rheumatic fever's association with poverty; iodine deficiency disease prevention through iodized salt; establishing the causal link between tobacco/asbestos and lung cancer (combined exposure has multiplicative effect: RR =

602/100,000 vs 11/100,000 in unexposed); understanding of HIV/AIDS (40 million infected by 2005); and response to SARS (8000 cases, 900 deaths in 12 countries in 2002-03).

Chapter 2: Measuring Health and Disease

Key Messages

- Measuring health and disease is fundamental to epidemiology.
- Various measures characterize overall population health.
- Population health status is not fully measured in many parts of the world.

Defining Health and Disease

WHO definition (1948): "Health is a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity." Practical definitions used in epidemiology tend to be simple ("disease present" or "disease absent"). Cut-off points for defining abnormality are operational definitions, not absolute thresholds. Diagnostic criteria are based on symptoms, signs, history and test results, and may change as knowledge increases or diagnostic techniques improve.

Measuring Disease Frequency

Population at Risk

People susceptible to a given disease, defined by demographic, geographic or environmental factors. E.g., only women should be included when calculating cervical cancer frequency.

Prevalence

P = Number of people with disease at specified time / Number of people in population at risk at specified time (x 10^n)

Prevalence is influenced by: severity of illness, duration of illness, number of new cases, migration patterns, and diagnostic improvements. Approximately: Prevalence = Incidence x Average duration of disease.

Incidence

I = Number of new events in a specified period / Number of persons exposed to risk during this period (x 10^n)

Incidence represents the rate of occurrence of new cases. The denominator is person-time at risk. Cumulative incidence = Number who get disease during period / Number free of disease at beginning of period. Attack rate is incidence during a disease outbreak in a narrowly-defined population over a short time period.

Case Fatality

Case fatality (%) = Number of deaths from disease in period / Number of diagnosed cases in same period x 100

Mortality Measures

Crude mortality rate = Deaths during period / Population at risk during period (x 10^n). Does not account for age, sex, race differences - requires standardization for valid comparisons.

Infant mortality rate = Deaths <1 year / Live births in same year x 1000. Sensitive to socioeconomic changes. Varies from 4/1000 (Japan) to 297/1000 (Sierra Leone males).

Maternal mortality rate = Maternal deaths / Live births (x 10^n). Ranges from ~3/100,000 in high-income countries to >1500/100,000 in low-income countries.

Life expectancy: Average years an individual of given age expects to live. Global increase from 46.5 years (1950-55) to 65.0 years (1995-2000). Reversals in sub-Saharan Africa (AIDS) and former Soviet Union (alcohol/tobacco).

Age-Standardized Rates

Age-standardized rates eliminate influence of different age distributions when comparing populations. Direct standardization applies disease rates to a standard population. Common standards: Segi world population, European standard population, WHO world standard population (2000-2025 projections).

Summary Measures of Population Health

DALYs (Disability-Adjusted Life Years) = YLL (Years of Lost Life) + YLD (Years Lost to Disability). One DALY = one lost year of "healthy" life. Combines premature mortality with disability impact. A death in infancy = 33 DALYs. Other measures: HALE (Healthy Life Expectancy), QALYs (Quality-Adjusted Life Years).

Comparing Disease Occurrence

Risk difference (excess risk) = Rate in exposed - Rate in unexposed. Attributable fraction (exposed) = (Rate in exposed - Rate in unexposed) / Rate in exposed x 100. Population attributable risk (PAR) = (Incidence in total population - Incidence in unexposed) / Incidence in total population.

Relative risk (risk ratio) = Risk in exposed / Risk in unexposed. Better indicator of association strength. RR of lung cancer in heavy smokers vs non-smokers is approximately 20.

Chapter 3: Types of Studies

Key Messages

- Choosing appropriate study design is crucial in epidemiological investigation.
- Each study design has strengths and weaknesses.
- Epidemiologists must consider all sources of bias and confounding.
- Ethical issues are important in epidemiology.

Observational Studies

Descriptive Studies

Description of health status based on routinely available data. Often the first step in investigation. Examine patterns by age, sex, ethnicity, time, and geography. Do not analyse links between exposure and effect.

Ecological Studies

Units of analysis are groups (populations), not individuals. Useful for generating hypotheses. Ecological fallacy: association at group level may not represent individual-level association. Time-series studies (very short time periods) minimize confounding as people serve as their own controls.

Cross-Sectional Studies

Measure prevalence at one point in time. Exposure and effect measured simultaneously. Relatively easy and inexpensive. Useful for fixed characteristics and assessing healthcare needs. Limited in proving causation due to uncertain time sequence.

Case-Control Studies

Compare exposure history in people WITH disease (cases) vs. people WITHOUT (controls). Longitudinal design looking backward from disease to possible cause. Efficient for rare diseases. Cannot determine absolute incidence. Measure of association: Odds Ratio (OR) = (a x d) / (b x c) from 2x2 table. OR approximates relative risk when disease is rare. Example: Thalidomide study - 41/46 cases exposed vs 0/300 controls.

Cohort Studies

Follow disease-free people classified by exposure status over time. Best information about causation; most direct measurement of risk. Can examine multiple outcomes. Expensive, require long follow-up. Types: Prospective, historical (retrospective), nested case-control. Example: British Doctors Study, Nurses' Health Study (121,700 women), Framingham Study (since 1948).

Experimental Studies

Randomized Controlled Trials (RCTs): Gold standard. Random allocation ensures group equivalence. Field Trials: Healthy people "in the field" (e.g., Salk polio vaccine trial - 1 million children). Community Trials: Treatment groups are communities (e.g., cardiovascular disease prevention). Limited by small number of communities and difficulty isolating from general social changes.

Potential Errors

Random Error

Sources: individual biological variation, sampling error, measurement error. Reduced by increasing sample size and stringent measurement protocols.

Systematic Error (Bias)

Selection bias: Systematic difference between selected participants and those not selected. Includes healthy worker effect, self-selection, loss to follow-up. Measurement (classification) bias: Inaccurate measurement of disease or exposure. Recall bias: Differential recall by cases vs. controls. Observer bias: Knowledge of exposure status influences measurements. Controlled by blinding (single-blind or double-blind studies).

Confounding

Occurs when an extraneous factor is associated with both disease and exposure being studied. Example: Coffee drinking and heart disease confounded by smoking. Control methods in design: randomization, restriction, matching. Control in analysis: stratification, statistical modeling (multivariate analysis).

Validity

Internal validity: Results correct for the group being studied. External validity (generalizability): Results apply to people not in the study. High reliability does not ensure validity (measurements may all be far from true value).

Chapter 4: Basic Biostatistics - Concepts and Tools

Key Messages

- Basic epidemiology requires knowledge of biostatistics.
- Good quality tables and graphs communicate data effectively.
- Confidence intervals are valuable estimation tools.
- Calculations can be complex, but underlying concepts are often simple.

Summarizing Data

Data types: Numerical (counts, measurements) and Categorical (classifications, ordinal ranks). Visualization tools: Pie charts, component band charts, spot maps, rate maps, bar charts, line graphs, frequency distributions, histograms. Normal distribution: ~68% within 1 SD of mean, ~95% within 2 SD.

Summary Numbers

Central tendency: Mean (average), Median (middle value - robust to outliers), Mode (most frequent). Variability: Variance = SS(x)/(n-1), Standard deviation = sqrt(variance), Standard error of mean = s/sqrt(n) - reflects how similar sample means would be if repeated.

Statistical Inference

Confidence Intervals

Provide reasonable bounds for population parameters based on sample data. For 95% CI: Lower/Upper bound = mean +/- t(0.975) x (s/sqrt(n)). Interpretation: 95% of such intervals would contain the true population mean. Narrower intervals are better; larger samples produce shorter intervals. Can be used to test hypotheses: reject null if hypothesized value falls outside CI.

Hypothesis Testing and P-values

Null hypothesis (H0) vs alternative (H1). Alpha level usually 0.05. P-value: probability of observing result as extreme or more, assuming H0 is true. Reject H0 when p < alpha. Two types of errors: Type I (alpha-error): rejecting H0 when true. Type II (beta-error): accepting H0 when false. Statistical power = 1 - beta (probability of detecting a true difference). Desirable power >= 0.80.

Basic Methods

t-tests: Compare means of two populations. Uses t-statistic with (n1+n2-2) degrees of freedom.
Chi-squared tests: Test association between categorical variables in contingency tables. Compare observed vs expected frequencies. Chi-sq = sum of (O-E)^2/E.
Correlation: Pearson coefficient r measures linear association (-1 to +1).
Regression: Linear (continuous outcome), Logistic (binary outcome - OR), Cox proportional hazards (time-to-event - RR/HR). R-squared measures proportion of variation explained.

Sample Size and Meta-Analysis

Sample size depends on: significance level, acceptable error, magnitude of effect, disease prevalence. Meta-analysis: Statistical synthesis of data from separate but comparable studies. Steps: formulate problem, identify studies, exclude poor studies, measure and combine results. Produces pooled estimate with narrow confidence interval from aggregated data.

Chapter 5: Causation in Epidemiology

Key Messages

- Study of causation is fundamental to epidemiology.
- There is seldom one single cause of a specific health outcome.
- Causal factors can be arranged into a hierarchy from proximal to distal.
- Criteria for causality: temporal relationship, plausibility, consistency, strength, dose-response, reversibility, study design.

The Concept of Cause

A cause must precede an outcome. Sufficient cause: inevitably produces outcome. Necessary cause: outcome cannot develop without it. Most diseases have multiple causes (multi-factorial). Removal of one component may prevent disease even without identifying all components.

Koch's postulates (organism present in every case, isolated in pure culture, causes disease in animals, recovered from animal) are useful but inadequate for many diseases, especially non-communicable ones.

Four types of causal factors: Predisposing (age, sex, genetics), Enabling/disabling (income, nutrition, housing), Precipitating (specific exposure), Reinforcing (repeated exposure, environment). Interaction: Combined effect greater than sum of individual effects (e.g., smoking + asbestos = 50x risk, not 15x).

Hierarchy of Causes

Proximal causes (precipitating factors) vs distal causes (enabling factors). Example: Inhaled tobacco smoke is proximal cause of lung cancer; low socio-economic status is distal cause. DPSEEA framework: Driving forces -> Pressure -> State -> Exposure -> Effect -> Action. In analysis, "causes of causes" require appropriate statistical methods.

Establishing Cause: Considerations for Causation

1. Temporal relationship (ESSENTIAL): Cause must precede effect.
2. Plausibility: Consistent with other knowledge and biological mechanisms.
3. Consistency: Similar results in different studies, settings, designs.
4. Strength: Larger risk ratios (RR>2 considered strong) more likely causal.
5. Dose-response relationship: Increased exposure = increased effect.
6. Reversibility: Removal of cause reduces disease risk.
7. Study design: RCTs strongest, then cohort, case-control, cross-sectional, ecological.
8. Judging evidence: Multiple lines of evidence strengthen conclusion.

Before assessing causality, must exclude chance, bias, and confounding. Systematic review and meta-analysis pool results for better overall estimates. Precautionary principle may be applied when evidence is incomplete but risk is high.

Chapter 6: Epidemiology and Prevention - Chronic NCDs

Key Messages

- Chronic NCDs are major public health challenges in most countries (36 million deaths/year, 61% of global deaths).
- Causes are generally known and cost-effective interventions available.
- Comprehensive approach required: population + individual strategies.
- Primary prevention is the best long-term strategy.

Scope of Prevention

Leading chronic diseases: CVD (17.5M deaths), cancer (7.5M), chronic respiratory disease (4M), diabetes (1.1M). 80% of NCD deaths occur in low- and middle-income countries. Without greater prevention, by 2030 MI, stroke and diabetes will account for 4/10 deaths in adults 35-64 in LMICs.

Death rates from CVD have fallen 70% since 1970s in Australia, Canada, Japan, UK, USA. A decline of 2% per annum over 10 years could avert 35 million untimely deaths. Social determinants of health (conditions in which people live and work) are fundamental upstream causes.

Levels of Prevention

Primordial Prevention

Avoid emergence of social, economic and cultural patterns contributing to elevated disease risk. Includes national nutrition programmes, physical activity promotion, tobacco control policies. Framework Convention on Tobacco Control (2006) - first WHO health treaty. Tobacco taxation effectively reduces consumption.

Primary Prevention

Reduce incidence by controlling causes and risk factors. Two strategies:
1. Population strategy: Shift entire population distribution (e.g., reduce mean cholesterol). Advantages: radical, large potential, behaviourally appropriate. Disadvantage: small individual benefit.
2. High-risk individual strategy: Focus on those above arbitrary cut-off. More efficient for individuals at greatest risk but contributes less to overall burden. Prevention paradox: Most cases come from middle of risk distribution, not the tail.

Best approach: Combine both strategies. Decisions on treatment should consider overall absolute risk, not just single risk factor levels.

Secondary Prevention

Early diagnosis and treatment to reduce disease consequences. Requires: safe/accurate detection method and effective intervention during early disease period. Example: Cervical cancer screening - demonstrated association between screening rates and mortality reduction in Canadian provinces.

Tertiary Prevention

Reduce progress/complications of established disease. Includes rehabilitation. Often difficult to separate from treatment. Essential for restoring ability to work.

Screening

Process of using tests to identify disease in apparently healthy people. Types: Mass, multiple/multiphasic, targeted, case-finding/opportunistic.

Criteria for screening programmes: Well-defined disorder, known prevalence, long natural history, simple/safe test, known test performance, cost-effective, available facilities, acceptable procedures, equitable access.

Test validity: Sensitivity = a/(a+c) [proportion of diseased correctly identified]. Specificity = d/(b+d) [proportion of healthy correctly identified]. Positive predictive value = a/(a+b) [depends critically on disease prevalence]. Balance needed: increasing sensitivity reduces specificity and vice versa.

Chapter 7: Communicable Diseases - Surveillance and Response

Key Messages

- New communicable diseases emerge and old ones re-emerge with social/environmental change.
- Communicable diseases remain a continuing threat (14.2 million deaths/year).
- Epidemiological methods enable surveillance, prevention and control of outbreaks.
- International Health Regulations aim to facilitate control of new epidemics.

Definitions and Burden

Communicable disease: caused by transmission of pathogenic agent to susceptible host. Transmission: direct (contact, droplets) or indirect (vehicles, vectors, airborne). Six causes account for ~80% of infectious disease deaths: acute respiratory infections (3.76M), HIV/AIDS (2.8M), diarrhoeal diseases (1.7M), tuberculosis (1.6M), malaria (1M), measles (0.8M).

Epidemic and Endemic Disease

Epidemic: cases in excess of normal expectation. Point-source epidemic: simultaneous exposure (rapid increase in hours). Propagated epidemic: person-to-person spread (slower initial rise). Endemic: stable pattern of relatively high prevalence/incidence in geographic area. Herd immunity: when sufficient population is immune, even susceptible individuals are protected.

Emerging/re-emerging infections: >30 previously unknown diseases in recent decades. HIV/AIDS greatest impact. Avian influenza (H5N1): 50% case-fatality rate in confirmed human cases. Influenza pandemics (1918, 1957, 1968) caused tens of millions of deaths.

Chain of Infection

Four components: Infectious agent, Transmission process, Host, Environment.

Agent characteristics: Pathogenicity (ability to produce disease), Virulence (severity), Infective dose, Reservoir, Source (carriers important - e.g., HIV asymptomatic transmission).

Host factors: Points of entry, incubation period (hours to years), resistance (natural/acquired). Immunization types: living modified agents, inactivated organisms, inactive toxins, polysaccharides.

Environment: Sanitation, temperature, air pollution, water quality, population density, poverty.

Investigation and Control

Steps: Preliminary investigation -> Identify/notify cases -> Collect/analyse data -> Manage and control -> Disseminate findings. Control measures directed at: source, spread, and protection of exposed people. HIV control: condom promotion, needle exchange, blood screening, PMTCT with antiretrovirals.

Surveillance

Ongoing systematic collection, analysis and interpretation of health data linked to timely dissemination. Key principle: include only conditions where surveillance can lead to prevention. Sources: mortality/morbidity reports, hospital records, lab diagnoses, outbreak reports, vaccine utilization. International Health Regulations (2005): Countries must notify WHO of public health emergencies of international concern, verify outbreaks, maintain core capacity for early warning and response.

Chapter 8: Clinical Epidemiology

Key Messages

- Clinical epidemiology applies epidemiological principles to clinical medicine.
- Rising health-care costs make clinical practice a common subject of research.
- Evidence-based guidelines have improved clinical outcomes.
- Effective treatments are not fully used; ineffective treatments still prescribed.

Definitions of Normality and Abnormality

Three approaches:
1. Normal as common: Arbitrary cut-off (e.g., 2 SD from mean classifies 2.5% as abnormal). No biological basis for most variables - risk is a graded process.
2. Abnormality associated with disease: Overlap between healthy and diseased distributions makes perfect separation impossible. Balance sensitivity and specificity.
3. Abnormal as treatable: Determined by evidence from RCTs showing treatment does more good than harm. Criteria change over time as new evidence emerges (e.g., hypertension treatment thresholds have decreased). Should consider absolute risk, not just single risk factor values.

Diagnostic Tests

True positive, true negative, false positive, false negative. Positive predictive value: probability of disease given positive test (depends critically on prevalence). Negative predictive value: probability of no disease given negative test. Even highly sensitive/specific tests may have low positive predictive value if disease prevalence is low.

Natural History, Prognosis, and Treatment

Natural history stages: pathological onset -> presymptomatic stage -> clinically obvious disease. Prognosis expressed as probability of future events. Measured by case-fatality or survival probability. Survival analysis: Kaplan-Meier curves display time-to-event data, accounting for censoring.

Efficacy: Treatment does more good than harm in compliant patients. Effectiveness: Does more good than harm in all patients offered treatment. Best measured by RCTs. Compliance (extent to which patients follow advice) affects real-world effectiveness.

Evidence-Based Guidelines and Prevention

Guidelines: systematically developed recommendations for appropriate health care. Many patients, even in high-income countries, do not receive best evidence-based treatment. In LMICs: 20% of CHD patients not receiving aspirin, 50% not on beta-blockers.

Prevention in clinical practice: Smoking cessation (cumulative lung cancer risk: stopping at age 30 reduces risk nearly to never-smoker level). Fixed-dose combination therapy for CVD prevention (aspirin + beta-blocker + ACE inhibitor + statin) expected to reduce recurrent MI risk by 75%.

Chapter 9: Environmental and Occupational Epidemiology

Key Messages

- Environment strongly influences disease causation and injuries.
- Exposures quantified as "dose" to establish dose-effect and dose-response relationships.
- Health impact assessments forecast likely health impact of interventions.
- Injury epidemiology identifies effective preventive actions.

Environment and Health

Environmental factors: psychological, biological, physical, accidental, chemical. 25-35% of global disease burden caused by environmental exposure. Major problems: unsafe water/sanitation, indoor air pollution (biomass fuels), urban air pollution. Children bear highest death toll (>4 million environmental deaths/year). Infant death rate from environmental causes is 12x higher in low- vs high-income countries.

Occupational epidemiology: usually adult, young/middle-aged, often male, relatively healthy (healthy worker effect). General population includes children, elderly, sick - more sensitive to environmental factors. Example: Lead effects occur at lower levels in children (100 ug/l) vs adults (400 ug/l for neurobehavioural changes).

Exposure and Dose

Exposure has two dimensions: level and duration. External dose = exposure duration x exposure level. Biological monitoring: measuring chemical concentration in body fluids/tissues (blood, urine, hair, nails, breast milk). Absorbed (internal) dose vs external dose.

Dose-effect relationship: Higher dose = more severe effect in individual (e.g., carbon monoxide: slight headache -> dizziness -> nausea -> unconsciousness -> death).

Dose-response relationship: Proportion of exposed group developing specific effect. S-shaped curve reflecting individual sensitivity variation. Examples: Noise exposure and hearing loss (Table 5.2), asbestos and lung cancer (linear at low doses).

Risk Assessment and Management

Risk assessment: Assessment of health risk of defined policy/intervention. Health impact assessment: Focused on specific population or exposure situation. Environmental health impact assessment steps: Identify hazards -> Analyse health effects -> Measure exposure levels -> Combine exposure data with dose-response relationships.

Risk management: Plan and implement actions to reduce health risks. Includes setting safety standards based on dose-response relationships, monitoring exposure, and ensuring preventive measures are effective.

Example: Traffic air pollution in Europe - deaths from air pollution exceed traffic crash deaths (France: 17,629 vs 8,919; Switzerland: 1,762 vs 597).

Injury Epidemiology

Traffic crashes: Dose-response between speed and injury frequency. Pedestrian fatality rises steeply with car impact speed (near 100% at 80+ km/h). Seat-belt use dramatically reduces injury at all speeds.

Violence: Major cause of death in young males. Brazil: homicide = 40% of deaths in males 15-24.

Suicide: Access to method is environmental factor. Example: Western Samoa - suicides increased dramatically after paraquat introduction, decreased with control measures.

Climate change: Emerging challenge - broad health effects from temperature extremes, malaria spread, malnutrition, water shortages. Requires new epidemiological approaches.

Chapter 10: Epidemiology, Health Policy and Planning

Key Messages

- Epidemiology informs development, implementation, and evaluation of health policy.
- Epidemiologists can be usefully involved in health policy issues.
- Techniques for assessing interventions need refining.
- Health planning is a cycle incorporating continual assessment of effectiveness.

Health Policy

Health policy: Framework for health-promoting actions covering social, economic, and environmental determinants. Provides shape, direction and consistency to decisions.

The influence of epidemiology is often mediated by public opinion. Major difficulty: making judgements when evidence is incomplete. Time-scale for application varies - decades for chronic diseases.

Ottawa Charter (1985): Health influenced by wide range of policy decisions beyond health sector. Bangkok Charter (2005): Health promotion depends on empowering all sectors. Key commitments: central to global development, core government responsibility, community focus, good corporate practice. All policies affect health (agriculture, advertising, transport).

Framework Convention on Tobacco Control

First WHO health treaty (2006). 142 countries ratified by end of 2006 (77% of world population). Addresses demand reduction and supply reduction. Response to globalization of tobacco epidemic. Over 50 years of epidemiological evidence preceded this policy achievement.

The Health Planning Cycle

1. Assess burden: Measure health status (mortality, morbidity, DALYs, risk factors via WHO STEPS).
2. Identify causes: Determine major preventable causes (often same risk factors across societies).
3. Measure effectiveness: How much does intervention reduce morbidity/mortality? Best evidence from RCTs.
4. Determine efficiency: Cost-effectiveness analysis ($/DALY averted) or cost-benefit analysis.
5. Implement interventions: Set specific quantified targets.
6. Monitor and evaluate: Continuous follow-up to ensure activities proceed as planned.

STEPwise framework: Step 1 (core interventions with existing resources) -> Step 2 (expanded with projected resources) -> Step 3 (optimal evidence-based interventions).

Assessing Efficiency

Cost-effectiveness examples (cost per DALY averted):
- Tobacco taxation: $3-50 (most cost-effective)
- Treatment of MI with inexpensive drugs: $10-25
- MI drugs + streptokinase: $600-750
- Lifelong CVD treatment with "polypill": $700-1000
- Surgery for high-risk cases: $25,000+

Oral rehydration therapy: $2-4 per life-year saved - excellent value, saves millions of lives. WHO-CHOICE provides regional databases on costs and cost-effectiveness of interventions.

Chapter 11: First Steps in Practical Epidemiology

Key Messages

- Interesting epidemiology career depends on willingness to learn about diseases and risk factors.
- Knowing how to select and appraise reading material is important for staying informed.
- Good research requires good questions, clear protocols, ethical approval, and publication.
- Many online resources are freely available (databases, tools, references, guidelines).

Critical Reading

Five types of research papers: natural history, geographic distribution, causes, treatment, diagnostic tests. Levels of evidence progress from expert opinion through case-series, cohort studies, RCTs to systematic reviews.

Questions to ask: What is the research question? Is it relevant? What type of study? Who is included/excluded? How were samples selected? What sources of bias exist? Is the sample large enough? Are methods well described? Are results consistent? Can Type I/II errors be ruled out? Is there clinical/public health significance?

Planning Research

Steps: Choose project -> Write protocol -> Get approval -> Do research -> Analyse data -> Disseminate results.

Protocol should include: Clear problem description and approach; justification of importance; population/setting/intervention description; study design details (sampling, numbers, variables, data collection, quality assurance, analysis plan); budget/timetable; ethical approval plan; publication plan.

Consensus guidelines for reporting: CONSORT (RCTs), MOOSE (observational meta-analyses), TREND (non-randomized interventions), STARD (diagnostic accuracy), QUOROM (systematic reviews).

Ethical Requirements

All epidemiological studies must be reviewed by ethical review committees. Key principles: Informed consent (free and voluntary, right to withdraw); Confidentiality (preserve personal information); Respect for human rights (balance individual vs group interests); Scientific integrity (avoid conflicts of interest, fabricated data).

Summary of Key Concepts and Formulas

Disease Frequency Measures

Prevalence = Cases at time point / Population at risk
Incidence = New cases in period / Person-time at risk
Cumulative Incidence = New cases / Population at risk at start
Case Fatality = Deaths from disease / Diagnosed cases x 100
Prevalence approx = Incidence x Duration

Measures of Association

Risk Difference = Rate(exposed) - Rate(unexposed)
Relative Risk = Rate(exposed) / Rate(unexposed)
Odds Ratio = (a x d) / (b x c) [approximates RR for rare diseases]
Attributable Fraction = (Rate exposed - Rate unexposed) / Rate exposed x 100
Population Attributable Risk = (Rate total - Rate unexposed) / Rate total

Study Design Hierarchy (Ability to Prove Causation)

1. Randomized Controlled Trials (strongest)
2. Cohort Studies (moderate)
3. Case-Control Studies (moderate)
4. Cross-Sectional Studies (weak)
5. Ecological Studies (weakest)

Screening Test Performance

Sensitivity = True positives / (True positives + False negatives)
Specificity = True negatives / (True negatives + False positives)
Positive Predictive Value = True positives / All positives
Negative Predictive Value = True negatives / All negatives

Levels of Prevention

Primordial: Prevent emergence of conditions leading to causation (population-wide policies)
Primary: Reduce incidence by controlling causes and risk factors
Secondary: Early detection and treatment to reduce prevalence
Tertiary: Reduce complications of established disease (rehabilitation)

Considerations for Causation (Modified Hill Criteria)

1. Temporal relationship (essential - cause must precede effect)
2. Plausibility (consistent with other knowledge)
3. Consistency (similar results in different studies)
4. Strength (size of relative risk)
5. Dose-response relationship
6. Reversibility (removal of cause reduces risk)
7. Study design (quality of evidence)
8. Judging evidence (multiple lines of evidence)

Key Statistical Concepts

Confidence Interval: Range likely to contain true population parameter (usually 95%)
P-value: Probability of result if null hypothesis is true
Power: Probability of detecting true difference (target >= 80%)
Type I error (alpha): False positive (reject true H0)
Type II error (beta): False negative (accept false H0)

Confounding: Extraneous variable associated with both exposure and outcome
Meta-analysis: Statistical synthesis pooling results from comparable studies

Global Health Priorities

Chronic NCDs: 36M deaths/year (61% global deaths). Leading: CVD 17.5M, Cancer 7.5M, Respiratory 4M, Diabetes 1.1M. 80% in LMICs.
Communicable diseases: 14.2M deaths/year. Leading: Respiratory 3.76M, HIV/AIDS 2.8M, Diarrhoeal 1.7M, TB 1.6M, Malaria 1M, Measles 0.8M.
DALYs combine mortality and disability to inform resource allocation.
Millennium Development Goals: Poverty, education, gender, child mortality, maternal mortality, communicable diseases, environmental sustainability, global partnership.

Appendix A: Detailed Epidemiological Methods

Outbreak Investigation Steps

A systematic approach to outbreak investigation involves multiple stages:
1. Verify the diagnosis and confirm the outbreak exists (compare with baseline rates)
2. Define the case and count cases using standard case definitions
3. Orient data in terms of time (epidemic curve), place (spot map), and person (demographics)
4. Determine who is at risk of becoming ill (denominator)
5. Develop hypothesis about cause, source, and mode of transmission
6. Test hypothesis with analytic epidemiology (cohort or case-control study)
7. Plan and implement control measures to prevent further spread
8. Communicate findings through written report and recommendations

Common source outbreaks show point-source epidemic curve (sharp rise, gradual decline). Propagated outbreaks show multiple peaks reflecting person-to-person transmission. Identifying the index case helps trace transmission chains.

Surveillance Systems in Detail

Active surveillance: Regular contact with reporting sources to seek out cases. Resource-intensive but complete.
Passive surveillance: Relies on health workers submitting reports. Less resource-intensive but underreports.
Sentinel surveillance: Selected sites report in detail. Provides high-quality data from limited locations.

Data sources for surveillance include: Mortality registration systems, hospital discharge data, disease notification systems, laboratory results, cancer registries, birth defect registries, population surveys (demographic, health, behavioral), pharmacy records, and school absenteeism data.

WHO International Health Regulations (2005): Require countries to notify WHO of potential public health emergencies of international concern within 24 hours. Decision instrument helps assess whether an event requires notification based on seriousness, unusualness, risk of international spread, and risk of trade/travel restrictions.

Bias in Epidemiological Studies

Selection Bias: Systematic error from how subjects are selected or from differential loss to follow-up.
- Non-response bias: Responders differ systematically from non-responders
- Healthy worker effect: Workers generally healthier than general population
- Berkson bias: Hospital patients not representative of general population
- Survival bias: Studying prevalent cases misses rapidly fatal cases

Information (Measurement) Bias: Systematic error in measuring exposure or disease.
- Recall bias: Cases remember exposures differently than controls
- Observer bias: Knowledge of exposure status affects disease assessment
- Hawthorne effect: Behavior changes when subjects know they are observed
- Misclassification bias: Errors in categorizing subjects (differential vs non-differential)

Confounding: Variable associated with both exposure and outcome that distorts the true relationship.
- Control methods: Randomization, restriction, matching, stratification, multivariate analysis
- A confounder must be: associated with exposure, independently associated with disease, not on the causal pathway between exposure and disease

Clinical Trials: Detailed Methodology

Phases of drug trials:
- Phase I: Safety testing in healthy volunteers (20-80 subjects), pharmacokinetics
- Phase II: Preliminary efficacy in patients with condition (100-300 subjects)
- Phase III: Definitive RCT comparing new treatment vs standard (1000-3000+ subjects)
- Phase IV: Post-marketing surveillance for rare adverse effects

Randomization methods: Simple (coin flip), block (ensures balance within blocks), stratified (balance within important subgroups), cluster (randomize groups not individuals).

Sample size determination depends on: Expected effect size, significance level (alpha, usually 0.05), statistical power (1-beta, usually 0.80), variability in outcome, dropout rate estimate.

Intention-to-treat (ITT) analysis: All subjects analyzed in originally assigned group regardless of compliance or crossover. Preserves benefits of randomization. Per-protocol analysis may show efficacy but overestimates real-world effectiveness.

Appendix B: Applied Epidemiology Case Studies

Cardiovascular Disease Prevention

The major modifiable risk factors for CVD: hypertension, tobacco smoking, high blood cholesterol, physical inactivity, obesity, diabetes, and unhealthy diet (high salt, low fruit/vegetable).

Population strategy: Shift entire distribution of risk factor downward. Even small reductions in mean population blood pressure produce large reductions in CVD events. A 2 mmHg reduction in mean diastolic BP reduces CHD by 6% and stroke by 15%.

High-risk strategy: Identify and treat individuals at highest risk. Screening programs identify hypertensives (>140/90 mmHg) and hypercholesterolemics (>6.5 mmol/L). Limitations: Treatment must be lifelong, doesn't address underlying causes, medicalization of large proportion of population.

North Karelia Project (Finland): Comprehensive community-based intervention reduced CHD mortality by 73% in males over 25 years. Combined population and high-risk approaches: dietary change (reduced saturated fat, increased vegetables), smoking cessation programs, hypertension detection and treatment. Demonstrated that population-wide prevention works.

Tobacco and Health: An Epidemiological Triumph

Historical milestones in tobacco epidemiology:
- 1950: Doll and Hill case-control study linking smoking to lung cancer
- 1954: British Doctors Study cohort begins (followed for 50 years)
- 1964: US Surgeon General report declaring smoking causes cancer
- 1971: UK ban on TV advertising of cigarettes
- 2003: WHO Framework Convention on Tobacco Control adopted

Current tobacco burden: 5.4 million deaths/year globally. Expected to rise to 8.3 million by 2030. 70% of these deaths will be in developing countries. Tobacco kills up to half of long-term users.

Effective interventions (evidence-based):
- Price increases through taxation (10% increase reduces consumption 4-8%)
- Comprehensive advertising bans (reduce consumption 6-7%)
- Smoke-free legislation (reduces MI hospitalization 15-20%)
- Warning labels (graphic labels more effective than text)
- Cessation support (pharmacotherapy + counseling doubles quit rates)

HIV/AIDS Epidemic: Epidemiological Response

Epidemiological features that aided understanding:
- Cluster identification among MSM in US (1981) through surveillance
- Case-control studies identified sexual transmission patterns
- Cohort studies of hemophiliacs confirmed blood-borne transmission
- Perinatal transmission studies led to prevention (AZT reduces 68%)

Global response informed by epidemiology:
- Surveillance: HIV sentinel surveillance in antenatal clinics provides prevalence estimates
- Risk factor studies: Identified core groups and behavioral determinants
- Intervention trials: Male circumcision RCTs showed 60% reduction in acquisition
- Treatment as prevention: ART reduces transmission by 96% (HPTN 052 trial)
- Current: 38M living with HIV, 25.4M on ART (2019). Epidemic stabilizing in many regions.

Nutritional Epidemiology

Measurement challenges: Dietary assessment is inherently difficult due to day-to-day variation, recall bias, social desirability bias, and changing food compositions.

Methods: 24-hour recall, food frequency questionnaire, weighed food records, biomarkers.
Each has tradeoffs between accuracy, cost, and participant burden.

Key findings from nutritional epidemiology:
- Mediterranean diet reduces CVD events by 30% (PREDIMED trial)
- High sodium intake linked to hypertension and stroke
- Folic acid supplementation prevents neural tube defects (70% reduction)
- Vitamin A supplementation reduces child mortality by 23% in deficient populations
- Iron deficiency: Most common nutritional disorder globally (2 billion affected)
- Obesity epidemic: BMI >30 now affects 13% of world population

Appendix C: Epidemiology in Practice - Key Applications

Maternal and Child Health

Maternal mortality ratio: Deaths per 100,000 live births. Global estimate: 287,000 maternal deaths/year.
99% occur in developing countries. Lifetime risk: 1 in 39 (sub-Saharan Africa) vs 1 in 3,800 (developed).

Major causes: Hemorrhage (27%), hypertensive disorders (14%), sepsis (11%), unsafe abortion (8%), obstructed labor (9%), other direct causes (10%), indirect causes (20%).

Under-5 mortality: 6.9 million deaths/year. Leading causes: pneumonia (18%), preterm birth complications (14%), diarrhoea (11%), birth asphyxia (9%), malaria (7%). Nearly half associated with undernutrition.

Epidemiological contribution: Identified key interventions - skilled attendance at birth, emergency obstetric care, family planning, immunization coverage, oral rehydration, exclusive breastfeeding, insecticide-treated nets. These could prevent 63% of child deaths.

Occupational Epidemiology

Workplace exposures affect health through multiple pathways: chemical (solvents, metals, dusts), physical (noise, radiation, temperature), biological (infectious agents), ergonomic (repetitive strain), and psychosocial (stress, shift work).

Study designs commonly used:
- Proportional mortality studies (compare cause-specific mortality patterns)
- Standardized mortality ratio (SMR) studies (compare to general population)
- Nested case-control studies within occupational cohorts
- Cross-sectional surveys of current workers (survivor bias concern)

Key principles: Healthy worker effect must be considered (employed people healthier than general population). Latency periods often long for occupational cancers (10-40 years). Historical exposure reconstruction is often necessary.

Examples: Asbestos and mesothelioma (exposure-response well established), vinyl chloride and liver angiosarcoma, silica and silicosis/lung cancer, lead and neurodevelopmental effects in children.

Mental Health Epidemiology

Depression: Leading cause of disability worldwide. Lifetime prevalence 10-20% in most populations. Risk factors: female sex, family history, adverse childhood experiences, social isolation, poverty.

Alcohol use disorders: 3.3 million deaths/year attributable to alcohol. Dose-response relationship with liver cirrhosis, cancers, and injuries. J-shaped curve for CVD (moderate drinking may be protective).

Suicide epidemiology: 800,000 deaths/year globally. Male:female ratio 3:1 in high-income countries. Access to means (firearms, pesticides) is modifiable risk factor. Restricting access to lethal means is evidence-based prevention.

Challenges in psychiatric epidemiology: Case definition depends on clinical assessment criteria, stigma affects reporting and help-seeking, cultural factors influence symptom expression, co-morbidity is common (substance use + depression), and biological markers are limited.

Epidemiology of Aging

Demographic transition: Population aging is a global phenomenon. By 2050, people >60 years will exceed 2 billion (22% of global population). Fastest growth in developing countries.

Health implications: Multimorbidity increases with age (average 3+ chronic conditions in >65). Disability-free life expectancy is key outcome measure. Compression of morbidity hypothesis: can disability be postponed closer to death?

Geriatric syndromes: Falls (30% of >65 fall annually, leading injury cause), dementia (47M globally, projected 131M by 2050), frailty, sarcopenia, polypharmacy.

Evidence-based interventions: Physical activity (reduces falls 23%, all-cause mortality 30%), cognitive stimulation (delays dementia onset), social engagement, medication review, vaccination (influenza, pneumococcal), and screening for functional decline.

Appendix D: Emerging Issues in Epidemiology

Genomic Epidemiology

Integration of genetic information with traditional epidemiology to understand disease etiology.

Genome-wide association studies (GWAS): Scan entire genome for variants associated with disease. Identified thousands of susceptibility loci but most explain small proportion of risk.

Gene-environment interaction: Most diseases result from interplay between genetic susceptibility and environmental exposures. Examples: BRCA1/2 and breast cancer (penetrance varies by lifestyle), NAT2 slow acetylator phenotype and bladder cancer risk from aromatic amines.

Pharmacogenomics: Genetic variation affects drug metabolism and response. CYP2D6 polymorphisms affect metabolism of 25% of all drugs. Personalized medicine promise vs population health perspective.

Ethical issues: Genetic discrimination, privacy of genetic information, population consent for biobank research, return of incidental findings.

Digital Epidemiology and Big Data

New data sources: Electronic health records, mobile phone data, social media, internet search queries, wearable devices, satellite imagery, genomic databases.

Applications: Real-time disease surveillance (syndromic surveillance from emergency department data), prediction modeling (machine learning for outbreak detection), pharmacovigilance (adverse drug event detection from EHR data), contact tracing (mobile phone proximity data during pandemics).

Challenges: Data quality and completeness, selection bias (digital divide), privacy and consent (secondary use of data), overfitting and spurious correlations, lack of traditional study design controls, reproducibility concerns.

Google Flu Trends cautionary tale: Initially promising real-time flu surveillance using search queries, but overestimated by 50% in 2012-2013 due to media-driven search behavior changes. Highlights need for traditional epidemiological rigor even with big data.

Climate Change and Health

Direct effects: Heat-related mortality (European heatwave 2003: 70,000 excess deaths), injuries from extreme weather events (floods, storms, wildfires).

Indirect effects: Vector-borne disease expansion (malaria, dengue range shifts), water-borne diseases from flooding, food insecurity from agricultural disruption, air quality deterioration (ozone, particulate matter), forced migration.

Epidemiological approaches: Time-series studies of temperature-mortality relationships, spatial analysis of disease vector distributions, scenario modeling for future disease burden.

Co-benefits of mitigation: Active transport (cycling/walking) improves physical activity, reduced air pollution prevents 4.2 million deaths/year, plant-based diets reduce both emissions and NCD risk. Health framing motivates climate action.

Global Health Security

Emerging infectious diseases: Over 75% are zoonotic in origin. Drivers: Deforestation, urbanization, intensive farming, global travel, climate change.

International Health Regulations core capacities required: surveillance, response, preparedness, risk communication, human resources, laboratory, points of entry, zoonotic events, food safety, chemical events, radiation emergencies.

Pandemic preparedness lessons: Early detection and transparent reporting are critical. Mathematical modeling informs intervention timing and intensity. Non-pharmaceutical interventions (social distancing, masking, hand hygiene) are first-line until pharmaceutical countermeasures are available. Equity in vaccine distribution remains a challenge.

One Health approach: Recognizes interconnection of human, animal, and environmental health. Interdisciplinary collaboration essential for preventing spillover events and antimicrobial resistance.